13. Write the reason for the anamolous behaviour of Nitrogen.
14. Write the molecular formula and structural formula for the following molecules.
    a) Nitric acid b) dinitrogen pentoxide
    c) phosphoric acid d) phosphine
15. Give the uses of argon.
16. Write the valence shell electronic configuration of group-15 elements.
17. Give two equations to illustrate the chemical behaviour of phosphine.
18. Give a reaction between nitric acid and a basic oxide.
19. What happens when PCl_5 is heated?
20. Suggest a reason why HF is a weak acid, whereas binary acids of the all other halogens are strong acids.
21. Deduce the oxidation number of oxygen in hypofluorous acid – HOF.
22. What type of hybridisation occur in
    a) BrF_5 b) BrF_3
23. Complete the following reactions.
    1. $NaCl + MnO_2 + H_2SO_4 \longrightarrow$
    2. $NaNO_2 + HCl \longrightarrow$
    3. $IO_3^- + I^- + H^+ \longrightarrow$
    4. $I_2 + S_2O_3^{2-} \longrightarrow$
    5. $P_4 + NaOH + H_2O \longrightarrow$
    6. $AgNO_3 + PH_3 \longrightarrow$
    7. $Mg + HNO_3 \longrightarrow$
    8. $KClO_3 \xrightarrow{\Delta}$
    9. $Cu + \overset{\text{Hot conc}}{H_2SO_4} \longrightarrow$
    10. $Sb + Cl_2 \longrightarrow$
    11. $HBr + H_2SO_4 \longrightarrow$
    12. $XeF_6 + H_2O \longrightarrow$
    13. $XeO_6^{4-} + Mn^{2+} + H^+ \longrightarrow$
    14. $XeOF_4 + SiO_2 \longrightarrow$
    15. $Xe + F_2 \xrightarrow[400^0C]{\text{Ni / 200 atm}}$

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